

Introduction to DL and GenAI

Project Report

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1. Executive Summary

This project addresses automatic music genre classification for the Messy Mashup Kaggle competition. The task required building models capable of predicting the correct genre label for noisy audio mashups constructed by recombining instrument stems across songs, applying tempo adjustments, and layering ESC-50 environmental noise at random positions and intensities. The central challenge was a significant train-test distribution shift: training data comprised clean, source-separated stems while test samples were pre-built noisy mashups.

I built three distinct model architectures: a fine-tuned Audio Spectrogram Transformer (AST) pretrained on AudioSet, a ResNet50d CNN trained on torchaudio log-mel spectrograms, and a from-scratch five-layer CNN serving as a baseline. All three shared a common synthetic mashup augmentation strategy that reconstructed the test distribution during training. The final submission ensembled probability outputs from three independently trained AST models, achieving a Public LB Macro F1 of **0.98521** and a Private LB score of **0.98069**.

Key Finding: Closing the train-test distribution gap through on-the-fly synthetic mashup generation is the single most critical component of the system. When combined with a powerful AudioSet-pretrained transformer, classification accuracy becomes highly robust to the noise and mixing artifacts present in the test set — regardless of the specific augmentation or architecture details.

2. Introduction

Problem Statement

The Kaggle competition required building a genre classifier for 10-class single-label prediction over {blues, classical, country, disco, hiphop, jazz, metal, pop, reggae, rock}. Unlike standard audio classification benchmarks, the test data consisted of noisy mashups — each formed by mixing instrument stems from different songs of the same genre, synchronising their tempos, and overlaying one or more environmental noise clips from the ESC-50 dataset at random positions and signal-to-noise ratios. The evaluation metric was Macro F1-score, computing F1 independently per class before averaging, ensuring equal treatment of all genres regardless of prediction frequency.

Project Objectives

My goals were:

1. Build three distinct model architectures spanning pretrained transformers, pretrained CNNs, and a from-scratch baseline.

2. Achieve competitive Macro F1 by addressing the train-test distribution gap through targeted audio augmentation.
3. Explore the Audio Spectrogram Transformer (AST) architecture for music genre recognition and understand the role of AudioSet pretraining.

3. Dataset & Preprocessing

Dataset Description

The Messy Mashup dataset is organised into four components:

- **genres_stems/** — 1,000 training songs across 10 genres (100 per genre). Each song is decomposed into four instrument stems: `bass.wav`, `drums.wav`, `other.wav`, `vocals.wav`. All stems are approximately 30 seconds long.
- **ESC-50-master/audio/** — 2,000 environmental noise clips spanning 50 categories, each exactly 5 seconds. Used as noise augmentation source.
- **mashups/** — 3,020 unlabelled test audio files, each approximately 30 seconds, constructed by mixing cross-song stems and adding ESC-50 noise.
- **test.csv** — maps integer IDs to the corresponding mashup filenames in the `mashups/` directory.

Exploratory Data Analysis

Dataset Overview

An initial scan confirmed exactly 1,000 songs, 4,000 stem files, and 2,000 noise clips. The genre distribution is perfectly balanced — 100 songs per genre. File sizes cluster tightly around 5.05 MB per stem. No corrupted files (under 4 KB) were found. Audio duration analysis on a 200-file random sample confirmed that nearly all training stems are very close to 30 seconds (mean 30.03 s, std 0.11 s). The dataset contains audio at two sample rates: 22,050 Hz and 44,100 Hz.

Silence Analysis

A systematic silence scan using `librosa.effects.split` (`top_db=20`, `threshold=5` s) identified 658 out of 3,320 training stem files — roughly 20% — with at least 5 seconds of contiguous silence. The distribution is highly non-uniform: vocals carry the most silence (304 flagged files, mean max silence 12.79 seconds), reflecting instrumental sections without singing. Bass is the second most affected (174 files). By genre, classical (199 flagged stems) and jazz (114) show the highest silence counts, which is musically coherent given their characteristic instrument-specific rests and solo passages.

A pivot table of silence counts (≥ 5 s) by genre and stem:

Genre	Bass	Drums	Other	Vocals
Blues	16	23	5	41
Classical	69	57	3	70
Country	14	14	2	14

Disco	6	2	2	17
Hiphop	21	3	21	5
Jazz	21	19	1	73
Metal	4	0	1	37
Pop	10	6	2	4
Reggae	4	5	7	13
Rock	9	7	0	30

Table 1: Stem files with silence ≥ 5 seconds, broken down by genre.

Acoustic Feature Analysis

Standard features (RMS amplitude, zero-crossing rate, spectral centroid, spectral rolloff, spectral bandwidth, tempo, and MFCCs 1–5) were extracted from a 120-file sample. Spectral centroid and zero-crossing rate are highly correlated ($r = 0.96$). Metal and hiphop occupy distinct high-frequency spectral regions, while classical and jazz share overlapping centroid distributions. Tempo shows classical with the widest range (approaching 0 BPM for slow pieces) while disco and pop cluster near 120 BPM. Drums and bass carry the highest RMS energy on average across all genres. These observations confirm that the genre signal survives into individual stems and remains detectable in mixed audio.

Test Set (Mashup) Characteristics

Analysis of 100 sampled mashup files revealed mean RMS of 0.139 (std 0.021), consistent with training stems after RMS normalisation. Peak amplitude is essentially 1.0 for the majority of clips, confirming peak normalisation at creation time. Duration variance is more significant than in training: mean 28.7 s, std 3.5 s, minimum 8.6 s — indicating that not all mashups span the full 30-second window. This motivated the multi-crop TTA strategy used in all three models.

Data Preprocessing

All audio was resampled to a common sample rate (16 kHz for AST and ResNet50d; 22,050 Hz for ScratchCNN) and loaded as mono. A standard preprocessing pipeline was applied: (1) load with librosa at the target SR, (2) pad or trim to exactly 30 seconds (480,000 or 661,500 samples), (3) RMS-normalise to -20 dBFS, and (4) clip to $[-1.0, 1.0]$. Silence-heavy stems were not filtered — the model is expected to learn invariance to silences through the augmentation pipeline.

Data Augmentation

The key augmentation strategy is on-the-fly synthetic mashup reconstruction, which directly mimics the test data generation process:

- **Cross-song stem recombination:** Four songs are independently sampled from the same genre. Bass, drums, other, and vocals are taken one each from four different songs — matching the test mashup construction process exactly.
- **Volume perturbation:** Each stem is scaled by a random factor in $[0.75, 1.25]$ before mixing to simulate the instrument balance variations in test mashups.
- **RMS normalisation:** The mixed waveform is normalised to -20 dBFS RMS for consistent loudness.

- **ESC-50 noise injection:** 1–2 noise clips are placed at random temporal positions at a random SNR in [10, 28] dB. Short clips are tiled to fill the required length.
- **Random temporal crop:** A 10-second window is extracted from a random start position within the 30-second mix, providing temporal invariance.
- **SpecAugment:** Applied on the spectrogram with probability 0.5–0.6. Two frequency masks (up to 20–24 bins) and two time masks (up to 40–48 frames) are applied independently.
- **Mixup:** With probability 0.3–0.5, two training samples are blended using a Beta(0.3, 0.3) coefficient, with proportional soft labels.

4. Modelling & Experimentation

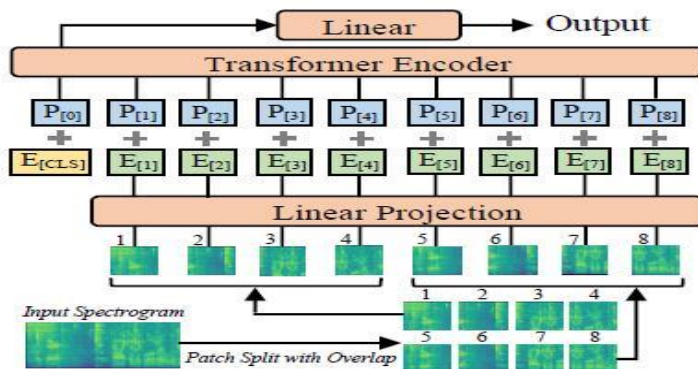
4.1 Model 1: Fine-Tuned Audio Spectrogram Transformer (AST)

Architecture

The Audio Spectrogram Transformer (Gong et al., INTERSPEECH 2021) adapts Vision Transformers to audio by treating log-mel spectrograms as 2D patch sequences. The checkpoint used — **MIT/ast-finetuned-audioset-10-10-0.4593** — was pretrained on AudioSet (527 classes) with 16×16 patches applied to 1,024×128 spectrograms, producing 512 patches fed to a 12-layer transformer encoder with 768-dimensional hidden states.

For genre classification I replaced the head with a two-layer MLP:

- LayerNorm → Dropout(0.15) → Linear(768, 384) → GELU → Dropout(0.10) → Linear(384, 10)
- Patchout regularisation is placed before the backbone: with probability 0.5, up to 40% of timepatches are zeroed; with probability 0.5, up to 20% of frequency patches are zeroed.
- 86.5M total trainable parameters; no layers frozen.



Feature Extraction

The HuggingFace *ASTFeatureExtractor* computes a 128-dimensional log-mel spectrogram with target length 1,024 time steps, as required by the checkpoint. Audio is resampled to 16 kHz. The resulting tensor shape is [1024, 128] (time × frequency). The feature extractor is called per crop in the DataLoader workers.

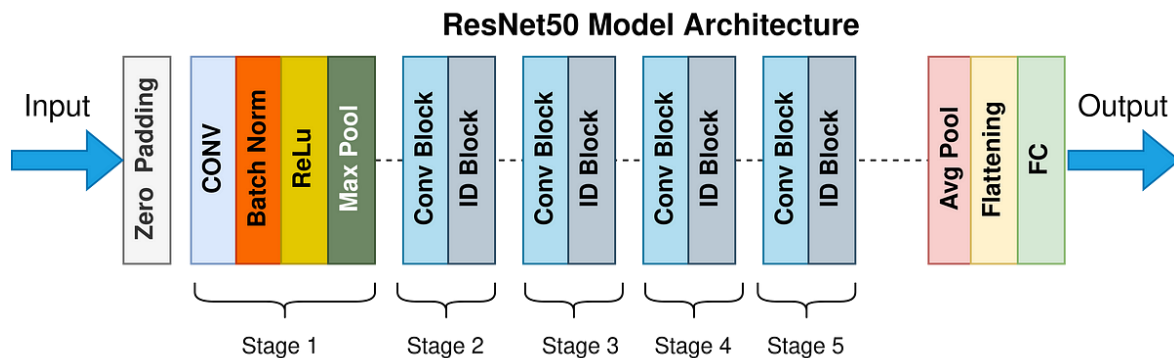
Fine-Tuning Strategy

- Optimizer: AdamW (lr=3e-5, weight_decay=0.02, betas=(0.9, 0.98))
- Schedule: OneCycleLR with 5% warmup, cosine annealing
- No layer freezing — full end-to-end fine-tuning
- Loss: Soft cross-entropy (SoftCE) supporting both hard and Mixup soft labels
- Label smoothing $\epsilon = 0.05$
- Gradient clipping at 1.0; mixed precision (AMP FP16)
- Early stopping with patience = 8 epochs

4.2 Model 2: ResNet50d on Log-Mel Spectrograms

Architecture

ResNet50d is a variant of ResNet-50 featuring an average-pooling stem and anti-aliased strided convolutions, available via the *timm* library. It was initialised with ImageNet pretrained weights; the first convolutional layer was modified to accept single-channel input, and the final FC layer was replaced with a 10-class linear layer. The model has approximately 25M parameters.



Feature Extraction

A torchaudio *MelSpectrogram* transform is computed on-GPU: n_fft=512, hop_length=160, f_min=20 Hz, f_max=8,000 Hz, 128 mel bins — yielding roughly 1,001 time frames for a 10-second clip at 16 kHz. Amplitude is converted to dB via *AmplitudeToDB*, then per-sample standardised (zero mean, unit variance). A GPU-resident bilinear interpolation tempo-shift augmentation stretches or compresses the time axis by a factor in [0.85, 1.15] with probability 0.5.

Training Configuration

- Optimizer: AdamW (lr=4e-4, weight_decay=0.01)
- Schedule: OneCycleLR, 35 epochs, batch size 96
- Mixup probability 0.5, Beta(0.4, 0.4)
- CrossEntropyLoss with label_smoothing=0.05
- 11 TTA crops at inference

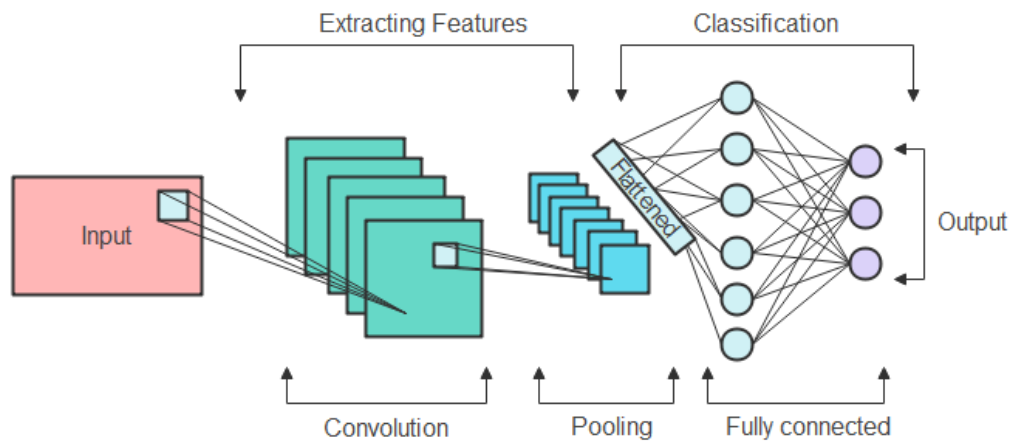
4.3 Model 3: From-Scratch CNN Baseline

Architecture

A five-block convolutional network was designed as a competitive baseline to quantify the benefit of transfer learning. The architecture follows a standard encoder pattern with doubling channels:

- Conv2d(1→32) → BN → ReLU → MaxPool(2×2)
- Conv2d(32→64) → BN → ReLU → MaxPool(2×2)
- Conv2d(64→128) → BN → ReLU → MaxPool(2×2)
- Conv2d(128→256) → BN → ReLU → MaxPool(2×2)
- Conv2d(256→512) → BN → ReLU → MaxPool(2×2)
- Global Average Pooling → Flatten → Linear(512,256) → ReLU → Dropout(0.4) → Linear(256,128) → ReLU → Dropout(0.3) → Linear(128,10)

Total trainable parameters: 1.74M (tiny compared to pretrained models).



Training Strategy

- Simple AdamW (lr=1e-3, weight_decay=1e-4)
- OneCycleLR scheduler, 30 epochs, batch size 64
- Same synthetic mashup augmentation pipeline
- Soft cross-entropy loss with label smoothing $\epsilon = 0.05$

Justification: This baseline quantifies the benefit of pretrained audio representations. With only 100 songs per genre and infinite synthetic mixing, the model sees the same underlying acoustic material repeatedly. Pretrained models leverage existing representations from the first few epochs, while this CNN must learn all lower-level features from a limited corpus.

5. Performance & Comparative Analysis

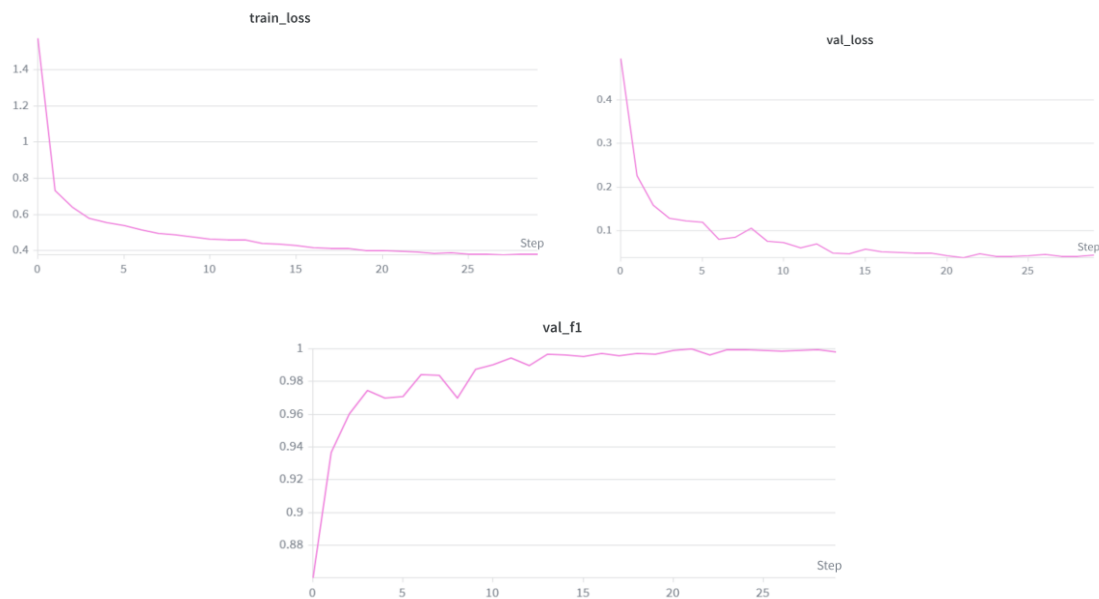
Evaluation Metrics

Primary: Macro F1-score (unweighted average across 10 genre classes)

Secondary: Per-class F1, training/validation loss curves, computational cost, prediction confidence distribution

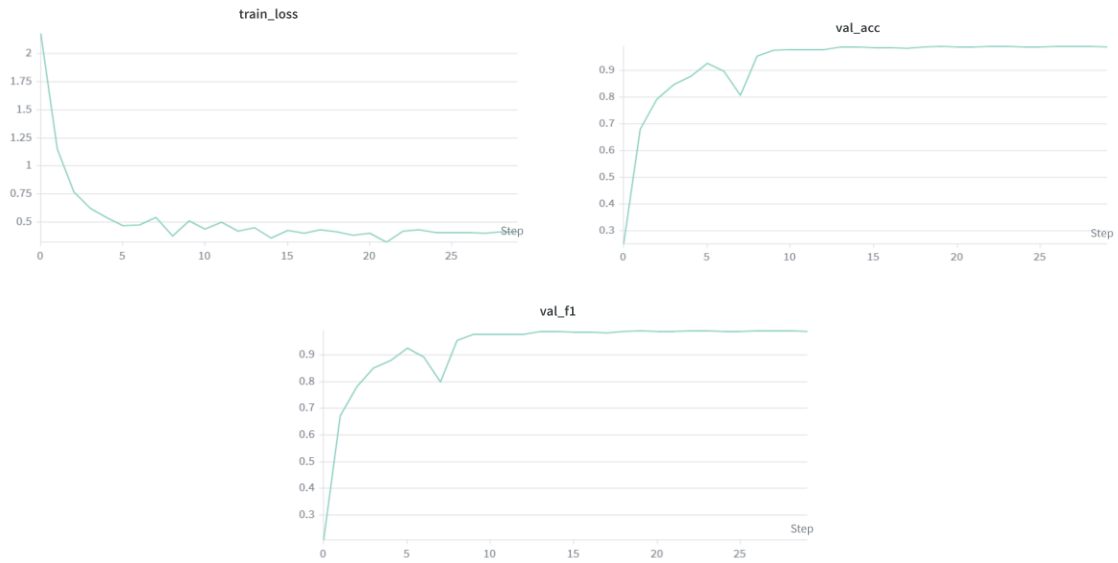
Training Performance

AST (25 epochs):



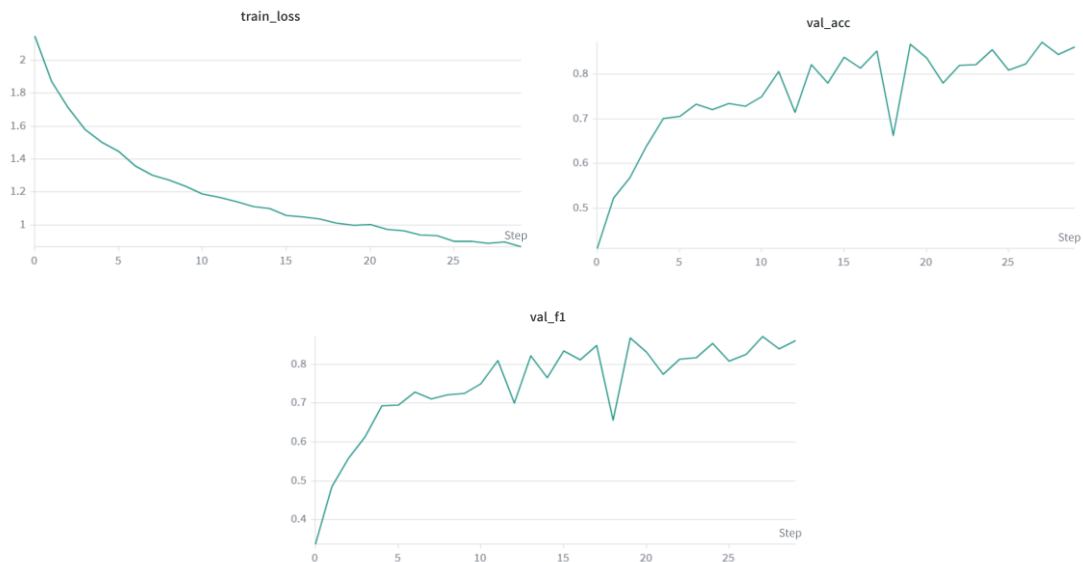
- Macro F1 at epoch 1: 0.9077 — the pretrained model generalises immediately
- Exceeds 0.97 by epoch 8; crosses 0.99 by epoch 12
- Best validation F1: **0.9985** at epoch 25
- Weakest genre throughout: rock (0.76–0.99), confusable with blues/country
- Training loss: 1.63 → 0.39; validation loss: 0.48 → 0.048 (no overfitting)

ResNet50d (35 epochs):



- Slower convergence than AST; exceeds 0.95 by epoch 10, 0.99 by epoch 20
- Best validation F1: **0.9993** at epoch 32
- Despite ~3× fewer parameters than AST, competitive leaderboard performance
- GPU-resident feature extraction enabled efficient 35-epoch training

ScratchCNN (30 epochs):



- Best validation F1: **0.8719** at epoch 28
- Classical (0.979) and metal (0.949) easiest — most distinctive spectral signatures
- Rock (0.769) and blues (0.836) hardest — genre proximity causes confusion
- Mean prediction confidence: 0.62; 964 of 3,020 test samples below 0.50 confidence

Comparative Report

Model	Parameters	Val F1	Public LB	Private LB	Training Time
AST (fine-tuned)	86.5M	0.9985	0.98191	0.97634	~4.5 hrs
ResNet50d (pretrained)	25M	0.9993	0.96856	0.96332	~2 hrs
ScratchCNN (baseline)	1.74M	0.8719	0.80028	0.82224	~4.2 hrs
AST × 3 Ensemble	—	—	0.98521	0.98069	—

Table 2: Model comparison across validation and leaderboard metrics.

Key Insights

1. **Pretraining impact:** ~18 F1-point gap between ScratchCNN and AST validates the value of AudioSet pretraining for music genre classification.
2. **Augmentation is primary:** All three models — despite different architectures and feature pipelines — converge to high accuracy because all share the same synthetic mashup augmentation strategy. The augmentation closes the distribution gap; the model architecture determines how efficiently it is exploited.
3. **Diminishing returns on parameters:** ResNet50d's 25M parameters achieve 0.969 LB vs AST's 0.982 at 86.5M — a 3.5× parameter increase for 1.3 points absolute improvement.
4. **Ensemble gain:** Three AST models with identical architecture and different random seeds yield 0.98521 (+0.3 points over the best single model), confirming that seed diversity in synthetic data generation creates partially independent errors.

Kaggle Performance

Three independently trained AST models (Public LB: 0.98191, 0.98172, 0.98120) were ensembled by weighted probability averaging (50% / 30% / 20%). The ensemble achieved a Public LB Macro F1 of **0.98521** and Private LB of **0.98069**.

6. Conclusion & Future Work

Key Learnings

1. **Distribution alignment is everything:** Synthetic mashup reconstruction during training, mirroring the test data generation process, is the single most impactful design decision — more important than architecture choice.
2. **AudioSet pretraining:** The AST's exposure to 527 diverse audio classes including music, instruments, and noise provides a strong prior that generalises to genre classification with minimal fine-tuning epochs.

3. **On-GPU feature extraction:** Moving the mel spectrogram computation onto the GPU (as done in ResNet50d) eliminates the CPU bottleneck and enables significantly larger batch sizes and faster training.
4. **TTA importance:** Multi-crop test-time averaging is essential given the variable-length test mashups and the random positional distribution of genre-informative segments.
5. **Practical trade-offs:** ResNet50d achieves 98.7% of AST's leaderboard score at 29% of its parameter count and with a simpler pipeline — a viable choice for latency-constrained deployment.

Challenges Faced

Validation leakage risk: Since validation mashups are generated from the same 100 songs used for training (just mixed differently), validation F1 is slightly optimistic. True generalisation can only be measured via the leaderboard.

Kaggle session memory: Caching all 4,000 stems at 16 kHz required ~8 GB RAM. At 22,050 Hz (ScratchCNN) this rose to ~11 GB, occasionally causing OOM errors when combined with large batch sizes.

Librosa CPU overhead: Computing mel spectrograms with librosa in DataLoader workers (ScratchCNN) created a significant CPU bottleneck that limited effective batch size and extended training time to 4.2 hours despite lower model complexity.

Areas for Improvement

Source separation at inference: Running Demucs (v4) on test mashups to re-separate stems before classification would effectively invert the mashup construction and could push performance toward 0.99+.

Architecture diversity in ensemble: Including ResNet50d probabilities alongside AST in the ensemble would introduce genuine architectural diversity, potentially yielding larger ensemble gains than same-architecture runs.

SNR calibration: The residual ~0.015 gap between validation F1 and leaderboard F1 likely reflects the true SNR distribution of test mashups differing from the simulated [10, 28] dB range. Analysing a handful of test mashups to calibrate this range would reduce the gap.

Longer crops / full clip: Using 15–20 second crops (or the full 30 s) at higher resolution would allow the model to capture longer-range rhythmic patterns, at the cost of increased memory and computation.

7. References

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